Thiago de Paula Oliveira

Statistician | Data Quality | Statistical Computing

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PROFESSIONAL SUMMARY

Statistician and statistical computing specialist with 14+ years of experience developing reliable analytical workflows, reusable R packages, dashboards, and reproducible reporting across agriculture, genetics, public health, and sports analytics. Strong background in improving data usability, analytical reliability, and decision-ready outputs through structured computational workflows, automated QC/ETL pipelines, documentation, and standardised analysis practices. Combines advanced statistical modelling with production-grade analytical tooling in R, C++, Bash, SQL, Docker, GitHub, and high-performance computing environments. Experienced in cross-functional collaboration, technical leadership, mentoring, and management responsibility for teams of up to six people.

Data quality | QC/ETL workflows | Reproducible workflows | Statistical modelling | R | SQL | Shiny dashboards | GitHub | Bash | C++ | Docker | High-performance computing

CORE CAPABILITIES

DATA STEWARDSHIP PRACTICE	Design and delivery of reproducible analytical workflows, automated QC/ETL pipelines, and structured computational environments that support reliability, traceability, and collaborative development.
DATA QUALITY AND USABILITY	Development of analytical dashboards, software packages, and decision-support tools that improve accessibility, consistency, and interpretability of complex data for researchers, analysts, and stakeholders.
STANDARDS AND DOCUMENTATION	Production of documented analytical pipelines, research software, and technical materials designed to support transparency, reuse, auditability, and maintainability of data analysis processes.
TECHNICAL LEADERSHIP	Experience leading cross-functional teams, mentoring colleagues, and managing teams of up to six people across analytical and computational projects.
STAKEHOLDER ENGAGEMENT	Strong written and verbal communication developed through consulting, interdisciplinary research, teaching, supervision, technical reporting, and collaboration with academic and industry partners.

PROFESSIONAL EXPERIENCE

2023–PRESENT

Consultant Statistician

AbacusBio

- Deliver statistical and analytical solutions in quantitative genetics, selection index development, and genomics for plant and animal breeding programmes.
- Lead cross-functional delivery of genetic evaluation pipelines, selection index tools, automated QC/ETL workflows, and decision dashboards for livestock, crop, and agri-tech partners.
- Develop and maintain reproducible analytical workflows, structured data pipelines, and practical decision-support tools for researchers and breeding teams.
- Maintain consistent reference datasets and analytical inputs across breeding workflows, supporting standardised interpretation of traits, identifiers, and evaluation outputs.
- Implement automated quality-control checks for pedigree, phenotype, and genomic datasets to improve reliability, consistency, and readiness for downstream analysis.

- Design traceable analytical pipelines linking raw breeding data, cleaned datasets, model inputs, and final genetic evaluation and selection index outputs.
- Work with production-grade code in R, C++, Bash, and SQL, using Docker-based environments to support reproducible and auditable delivery.
- Translate complex quantitative modelling into practical, transparent, and decision-ready outputs for applied breeding programmes.

2020–2023 **Research Fellow**
The Roslin Institute, University of Edinburgh

- Conducted research on quantitative genetics and genomic selection in plant breeding systems under the TRAIN@Ed fellowship.
- Developed statistical models and computational workflows for analysis of large-scale breeding datasets.
- Used high-performance computing environments to support large genomic data analyses.
- Produced research outputs, software contributions, and technical reports within collaborative interdisciplinary projects.
- Defined structured metadata for simulation and breeding datasets, including trait definitions, genomic variables, and environmental covariates, to ensure consistent interpretation and use across modelling and analytical workflows.

2020 **Postdoctoral Researcher in Biostatistics**
National University of Ireland Galway

- Worked on modelling approaches for early detection of secondary COVID-19 infection waves.
- Applied statistical modelling to public-health surveillance data.
- Delivered applied research outputs supporting epidemiological monitoring.

2020 **Postdoctoral Researcher in Biostatistics**
National University of Ireland Galway

- Contributed to interdisciplinary research projects within the Aspire Academy programme.
- Applied statistical analysis to research questions linking data science and applied performance contexts.

2019 **Postdoctoral Researcher in Biostatistics**
National University of Ireland Galway

- Developed statistical modelling approaches for optimisation of athlete performance.
- Collaborated with Orreco and the Insight Centre for Data Analytics on applied sport science analytics.
- Produced applied modelling outputs to support decision making in elite sport environments.

2018–2019 **Postdoctoral Researcher in Statistics**
University of São Paulo

- Led development of the `lcc` R package for estimation of longitudinal concordance correlation.
- Developed reusable statistical software supporting consistent and efficient analysis workflows.

- Combined methodological statistical research with production-quality software implementation.

2017–2019 | **Assistant Professor**
University of São Paulo
- Delivered teaching and supervision in statistics and quantitative methods.
- Led and supervised research projects involving applied statistical modelling and data analysis.
- Managed teams of students and researchers working on statistical and computational projects.

EDUCATION

2014–2018 | **PhD in Statistics**
University of São Paulo
Research focused on statistical modelling and methodological development for applied data analysis.

2012–2014 | **MSc in Statistics**
University of São Paulo
Graduate training in statistical modelling, inference, and computational statistics.

2008–2012 | **BSc in Agricultural Engineering**
University of São Paulo
Undergraduate training in agricultural systems, quantitative methods, and applied analysis.

SOFTWARE AND TECHNICAL TOOLS

PROGRAMMING	R, SQL, Bash, C++
DATA AND ANALYTICS	Statistical modelling, quantitative genetics, experimental design, data analysis, QC/ETL workflows
COMPUTING	Linux, Windows, and macOS environments; Docker; high-performance computing
REPRODUCIBLE RESEARCH	GitHub, LaTeX, R package development, auditable analytical workflows

SELECTED RESEARCH OUTPUTS

Oliveira, T.d.P.; Obšteter, J.; Pocrnic, I.; Heslot, N.; Gorjanc, G. A method for partitioning trends in genetic mean and variance to understand breeding practices. *Genetics Selection Evolution*, 2023. DOI: 10.1186/s12711-023-00804-3.

Richardson, C.; Amer, P.; Post, M.; Oliveira, T.d.P.; Grant, K.; Crowley, J.; Quinton, C.; Miglior, F.; Fleming, A.; Baes, C. F.; Malchiodi, F. Breeding for sustainability: Development of an index to reduce greenhouse gas in dairy cattle. *Animal*, 2025. DOI: 10.1016/j.animal.2025.101491.

Oliveira, T.d.P.; Newell, J. A hierarchical approach for evaluating athlete performance with an application in elite basketball. *Scientific Reports*, 2024. DOI: 10.1038/s41598-024-51232-2.

Das, K.; Oliveira, T.d.P.; Newell, J. Comparison of markerless and marker-based motion capture systems using 95% functional limits of agreement in a linear mixed-effects modelling framework. *Scientific Reports*, 2023. DOI: 10.1038/s41598-023-49360-2.

Oliveira, T.P.; Moral, R.A. Global short-term forecasting of COVID-19 cases. *Scientific Reports*, 2021. DOI: 10.1038/s41598-021-87230-x.